THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 21	Squares have an odd number of factors; count the integers whose squares lie in the range.
2	**(d)** 11	$n = 11$ gives 121, which is composite.
3	**(a)** 3	$n = 3$ gives 203, which is composite.
4	**(a)** 38	The middle number is the average, 36.
5	**(b)** 1	$7^2 = 49$ and $49 = 3 \times 16 + 1$.
6	**(a)** 8	$10a + b - (a + b) = 9a$, so the quotient is the tens digit.
7	**(c)** 41	$x^2 = 9^2 + 40^2 = 1681$.
8	**(c)** 12	Compare numbers of the form $p^a q^b$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The converse is 'If a number is even, then it is divisible by 6'; 4 is a counterexample, which is exactly R.
10	**(a)** Both A and R are true and R is the correct explanation of A.	A single counterexample, the number 2, disproves the statement.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2 = 20^2 + 21^2 = 841$ (1); $x = 29$ (1).	1+1
12	Take the LCM 12; 12 is divisible by 4 and 6 (1); But 12 is not divisible by 24 (1).	1+1
13	$11^2 - 1 = 120 = 8 \times 15$ (1); Converse: if $n^2 - 1$ is a multiple of 8, then n is odd (1).	1+1
14	$x^2 = 11^2 + 60^2 = 3721$ (1); $x = 61$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$8^2 + 9^2 = 64 + 81 = 145$, odd; General: $n^2 + (n+1)^2 = 2n^2 + 2n + 1$ (1); $= 2(n^2 + n) + 1$ (1); Odd (1).	1+1+1
16	$x = 12$ is a multiple of 3; $y = 60 = 12 \times 5$ (1); General: $x = ap$, $y = xq = a(pq)$ (1); A multiple of the given number (1).	1+1+1
17	$21^2 = 441$ and $20^2 = 400$; Sum $= 841$ (1); $29^2 = 841$ equals the sum (1); So the triangle is right-angled (1).	1+1+1
	OR: $12^2 + 13^2 = 144 + 169 = 313$, odd; General: $n^2 + (n+1)^2 = 2n^2 + 2n + 1$ (1); $= 2(n^2 + n) + 1$ (1); Odd (1).	1+1+1
18	$x = 14$ is a multiple of 7; $y = 84 = 14 \times 6$ (1); General: $x = ap$, $y = xq = a(pq)$ (1); A multiple of the given number (1).	1+1+1
	OR: $35^2 = 1225$ and $12^2 = 144$; Sum $= 1369$ (1); $37^2 = 1369$ equals the sum (1); So the triangle is right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Hypothesis: a number is a multiple of 8; conclusion: it is a multiple of 4 (1). (ii) 'If a number is a multiple of 4, then it is a multiple of 8' (1). (iii) True, since $8k = 4(2k)$ (1). (iv) False (1); 12 is a multiple of 4 but not of 8 (1).	1+1+1+2
20	$9 \times 10 \times 11 = 990 = 165 \times 6$ (1); Among three consecutive integers one is a multiple of 3 (1); At least one is even (1); Both factors 2 and 3 present, so divisible by 6 (2).	1+1+1+2
	OR: $13^2 = 169 = 21 \times 8 + 1$ (1); Let the odd integer be $2m + 1$ (1); Square $= 4m^2 + 4m + 1 = 4m(m+1) + 1$ (1); $m(m+1)$ is even, so $4m(m+1)$ is a multiple of 8 (1); Hence $8k + 1$ (1).	1+1+1+1+1

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.
- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 41	$n = 41$ gives 1681, which is composite.
2	**(c)** 48	The middle number is the average, 45.
3	**(c)** 1	$11^2 = 121$ and $121 = 3 \times 40 + 1$.
4	**(b)** 5	$10a + b - (a + b) = 9a$, so the quotient is the tens digit.
5	**(c)** 53	$x^2 = 28^2 + 45^2 = 2809$.
6	**(d)** 36	Compare numbers of the form $p^a q^b$.
7	**(b)** 34	Squares have an odd number of factors; count the integers whose squares lie in the range.
8	**(c)** 4	$n = 4$ gives 119, which is composite.
9	**(c)** A is true but R is false.	A is true; R is false because the converse 'If $x^2 = 9$, then $x = 3$' fails for $x = -3$.
10	**(d)** A is false but R is true.	A is false (examples do not prove a general statement); R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take the LCM 18; 18 is divisible by 6 and 9 (1); But 18 is not divisible by 54 (1).	1+1
12	$15^2 - 1 = 224 = 8 \times 28$ (1); Converse: if $n^2 - 1$ is a multiple of 8, then n is odd (1).	1+1
13	$x^2 = 13^2 + 84^2 = 7225$ (1); $x = 85$ (1).	1+1
14	Take the LCM 30; 30 is divisible by 10 and 15 (1); But 30 is not divisible by 150 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$15^2 + 16^2 = 225 + 256 = 481$, odd; General: $n^2 + (n+1)^2 = 2n^2 + 2n + 1$ (1); $= 2(n^2 + n) + 1$ (1); Odd (1).	1+1+1
16	$x = 30$ is a multiple of 5; $y = 90 = 30 \times 3$ (1); General: $x = ap$, $y = xq = a(pq)$ (1); A multiple of the given number (1).	1+1+1
17	$33^2 = 1089$ and $56^2 = 3136$; Sum $= 4225$ (1); $65^2 = 4225$ equals the sum (1); So the triangle is right-angled (1).	1+1+1
	OR: $19^2 + 20^2 = 361 + 400 = 761$, odd; General: $n^2 + (n+1)^2 = 2n^2 + 2n + 1$ (1); $= 2(n^2 + n) + 1$ (1); Odd (1).	1+1+1
18	$x = 18$ is a multiple of 9; $y = 126 = 18 \times 7$ (1); General: $x = ap$, $y = xq = a(pq)$ (1); A multiple of the given number (1).	1+1+1
	OR: $55^2 = 3025$ and $48^2 = 2304$; Sum $= 5329$ (1); $73^2 = 5329$ equals the sum (1); So the triangle is right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) False (1). (ii) $2 + 3 = 5$ is odd (1). (iii) For $n = 4$, $2n + 1 = 9 = 3 \times 3$ is not prime (1). (iv) Let the odd numbers be $2a + 1$ and $2b + 1$; the product is $4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$ (1), which is of the form $2m + 1$, so it is odd (1).	1+1+1+2
20	$17 \times 18 \times 19 = 5814 = 969 \times 6$ (1); Among three consecutive integers one is a multiple of 3 (1); At least one is even (1); Both factors 2 and 3 present, so divisible by 6 (2).	1+1+1+2
	OR: $21^2 = 441 = 55 \times 8 + 1$ (1); Let the odd integer be $2m + 1$ (1); Square $= 4m^2 + 4m + 1 = 4m(m+1) + 1$ (1); $m(m+1)$ is even, so $4m(m+1)$ is a multiple of 8 (1); Hence $8k + 1$ (1).	1+1+1+1+1

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.
- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1	$14^2 = 196$ and $196 = 3 \times 65 + 1$.
2	**(a)** 9	$10a + b - (a + b) = 9a$, so the quotient is the tens digit.
3	**(d)** 85	$x^2 = 36^2 + 77^2 = 7225$.
4	**(a)** 48	Compare numbers of the form $p^a q^b$.
5	**(d)** 40	Squares have an odd number of factors; count the integers whose squares lie in the range.
6	**(c)** 3	$n = 3$ gives 143, which is composite.
7	**(c)** 3	$n = 3$ gives 203, which is composite.
8	**(b)** 67	The middle number is the average, 62.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$n = 1$ is a counterexample, so the statement is false; R states that counterexample.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A parallelogram that is not a rectangle has equal opposite sides, so the converse is false; R is true but only defines the converse and does not show it is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$19^2 - 1 = 360 = 8 \times 45$ (1); Converse: if $n^2 - 1$ is a multiple of 8, then n is odd (1).	1+1
12	$x^2 = 33^2 + 56^2 = 4225$ (1); $x = 65$ (1).	1+1
13	Take the LCM 24; 24 is divisible by 6 and 8 (1); But 24 is not divisible by 48 (1).	1+1
14	$25^2 - 1 = 624 = 8 \times 78$ (1); Converse: if $n^2 - 1$ is a multiple of 8, then n is odd (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$24^2 + 25^2 = 576 + 625 = 1201$, odd; General: $n^2 + (n+1)^2 = 2n^2 + 2n + 1$ (1); $= 2(n^2 + n) + 1$ (1); Odd (1).	1+1+1
16	$x = 20$ is a multiple of 4; $y = 180 = 20 \times 9$ (1); General: $x = ap$, $y = xq = a(pq)$ (1); A multiple of the given number (1).	1+1+1
17	$77^2 = 5929$ and $36^2 = 1296$; Sum $= 7225$ (1); $85^2 = 7225$ equals the sum (1); So the triangle is right-angled (1).	1+1+1
	OR: $31^2 + 32^2 = 961 + 1024 = 1985$, odd; General: $n^2 + (n+1)^2 = 2n^2 + 2n + 1$ (1); $= 2(n^2 + n) + 1$ (1); Odd (1).	1+1+1
18	$x = 18$ is a multiple of 6; $y = 144 = 18 \times 8$ (1); General: $x = ap$, $y = xq = a(pq)$ (1); A multiple of the given number (1).	1+1+1
	OR: $51^2 = 2601$ and $140^2 = 19600$; Sum $= 22201$ (1); $149^2 = 22201$ equals the sum (1); So the triangle is right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 'If all the sides of a quadrilateral are equal, then it is a square' (1). (ii) True, because every square has four equal sides (1). (iii) False (1). (iv) A rhombus with angles 60° and 120° has all four sides equal (1) but its angles are not right angles, so it is not a square (1).	1+1+1+2
20	$28 \times 29 \times 30 = 24360 = 4060 \times 6$ (1); Among three consecutive integers one is a multiple of 3 (1); At least one is even (1); Both factors 2 and 3 present, so divisible by 6 (2).	1+1+1+2
	OR: $33^2 = 1089 = 136 \times 8 + 1$ (1); Let the odd integer be $2m + 1$ (1); Square $= 4m^2 + 4m + 1 = 4m(m+1) + 1$ (1); $m(m+1)$ is even, so $4m(m+1)$ is a multiple of 8 (1); Hence $8k + 1$ (1).	1+1+1+1+1

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